□□□□□□□□□□□□□□ □□□□□□□□
□□□□□□□□ □□□□□□□□□□□
□□□□□□□ □□□□□□□□□□□ □□ □□□□□□□□
□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□
□□□□ AP-ARABIC-2026-v1.2

□□□□□□□ □□□□□□□□□□□□□□□□□□□□□□□

□□□□□ □□□□

[illegible]

[illegible]

- [illegible]
- [illegible]
- [illegible]

[illegible]

[illegible] structured/schema/. [illegible] .

[illegible] . [illegible] structured/.

[illegible] . [illegible] Geeza Pro [illegible] هيئة الزكاة والضريبة والجمارك. [illegible] monospace [illegible] . [illegible] structured/enums.yaml.

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

- [illegible]
- [illegible] 3CPJR5CQ…K10.pdf [illegible] B1STE8RH…SJ10.pdf [illegible] اشعارصدورفاتورة [illegible]
- [illegible]
- [illegible]

[illegible] source/Saudi…OTP… [illegible]

□□□ □□□□□□□□ □□□□ □□□□□□ □□□□□□ □□□□
□□□□□ □□□□□□ □□□□□ □□□□□□□□□□ □□□□ □□□□
□□□ □□□□ □□ □□□ □□□□□□□ □□□ □□ □□□□□□□□□□□□.

□.□ □□□□□

□□□ □□□□□□□□□□□□□□ □□□□□□□ □□□□□ □□□□
□□□□□□ □□□□ □□□□ □ □□□□□□ □□□□□□□ □□ □□□
□□□□□□ □□□□□□□□ □□ source/□

□. □□□□□□□□□□□□ □□□□□□□□□□. □□□□□ □□□□□□□ □□
□□□□□□□□□□ □□□□□□□□ □□□□□□□□ □ □□□□□□
□□□□□□□□□□□ □□ structured/schema/invoice.schema.json
□□□□□□□□ □□.

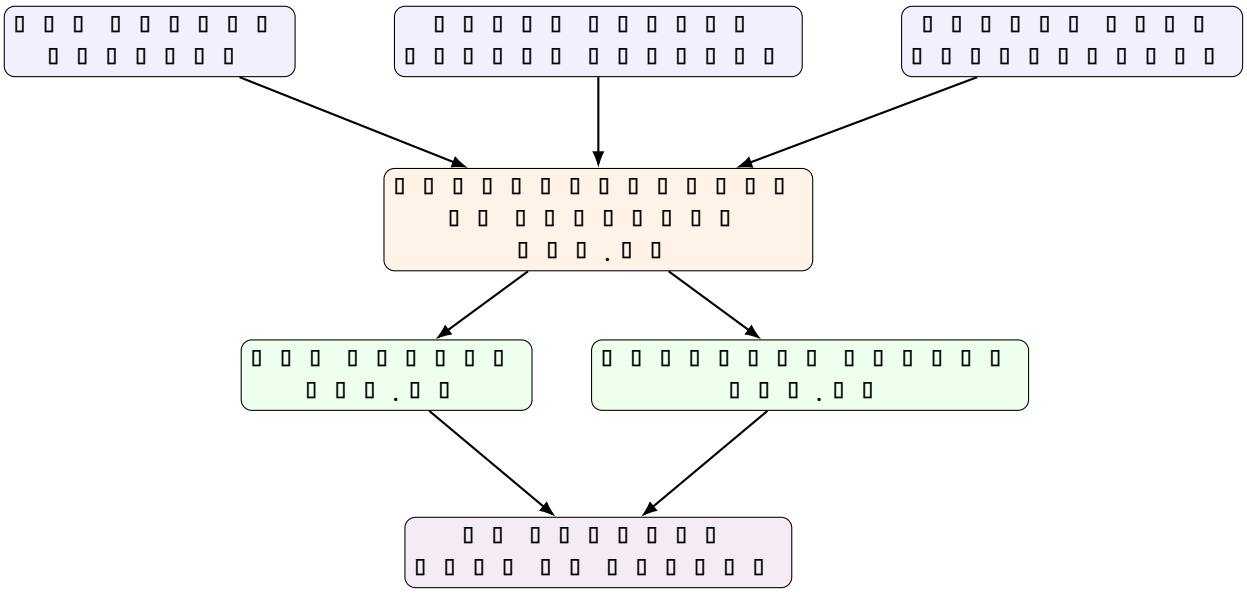
□. □□□□□□□□□□□□□□□ □□□□□□□□□□. □□□□□ □□□
□□□□□ □□ □□□□□□□ □□□□□□□□ □ □□□□ □□□□□□
□□□□ □□□□□□ □ □□□□ □□□□□□□□□□□□ □□□□□□□□
□□□□□□□□□□□ □□ structured/schema/vat-checklist.schema.json
□□□□□□□□ □□□□ □□□□□ □□□□ □□□□□ □□□□ □□□□□□□
□□□□ □□□□ □□□□□□□□□□□□□□□□□□□□□□□ □□□□□□□
□□ □□□□ □□□□□□□.

□. □□□□□□□□□□□□□□□□□□□□

[illegible]

[illegible]	[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]	3CPJR5CQ…K10.pdf B1STE8RH…SJ10.pdf
[illegible]	[illegible]	[illegible]	اشعار صدور فاتورة [illegible]
[illegible]	[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]	[illegible]

[illegible]



□□□□□□ □.□□ □□□□□□ □□□□□□□. □□□□□□□

□□□□□□□ □□□□ □□□□ □□□□□□□.

□□□□□ □.□□ □□□□□□□□□□□□□□□ □□□□□□□□□□ □□□□□
□□□□□□□□□□ structured/enums.yaml□.

□□□□	□□□□	□□□□
□□□□□□□□□□□	□□□□□□□□□	□□□ □□□□□□□□ □□□□□□ □□□□ □□□□□□□□ □□□□ □□□□□□.
□□□□□□	□□□□□□□□□	□□□□□□□ □□□□□□□□ □□□□ □□□□□□ □□□□□□.
□□□□□□	□□□□□□□□□	□□□□□□□□ □□□□□□□□□□□□□□ □□□□□□ □□□□□□.
□□□□□□□□□□	□□□□□□□□□	□□□□□ □□□□□□□□□□ □□□□ □□□ □□□ □□□□ □□□□.
□□□□	□□□□□□□□□	□□□□ □□□□ ≥□ □□□□□□ □□□□□□□□ □□□□□□□□□□□□□□□□□□□□□□□□□□□□.

[illegible]

[illegible] structured/

[illegible] structured/schema/. [illegible] structured/validate.py [illegible] $ref [illegible].

[illegible]

[illegible] structured/invoices/eg-eta-3CPJR5CQ4PNNN9A4XHBB370K10.json.

[illegible] structured/enums.yaml [illegible] ^[0-9A-Z]{26}$ [illegible] status [illegible]

□ structured/.

□ □ □ □	□ □ □ □ □ □ □ □
schema/invoice.schema.json	□ □ □ □ □ □ □ □ □ □ □ □ □ □ □ □ □ □ □

□ □ □ □

[illegible]

[illegible]

[illegible] payment-instruction.schema.json. [illegible] structured/payment-instructions/[illegible] [illegible]

[illegible] payment-instruction.schema.json.

[illegible]	[illegible]	[illegible]	[illegible]
id	[illegible]	✓	[illegible] payment-instructions/.
metadata	[illegible]	✓	[illegible]
payer	[illegible]	✓	[illegible]
payee	[illegible]	✓	[illegible]
amount	[illegible]	✓	value [illegible] currency [illegible]
payment_method	[illegible]	✓	[illegible] bank_transfer[illegible] check[illegible] card[illegible] cash[illegible] other.
reference	[illegible]	[illegible]	[illegible] invoice_id[illegible] description[illegible]
approval_chain	[illegible]	[illegible]	[illegible]
due_date	[illegible]	✓	[illegible]
status	[illegible]	✓	[illegible] draft[illegible] approved[illegible] emitted[illegible] settled[illegible] cancelled.

[illegible]

[illegible]

[illegible]

[illegible]	[illegible]	[illegible]	[illegible]
period	[illegible]	✓	start[illegible] end[illegible] label.
checklist[]	[illegible]	✓	[illegible] [illegible] …n[illegible] [illegible] result ∈ [illegible].
gl_coding	[illegible]	✓	[illegible]
supplier	[illegible]	✓	supplier_type ∈ [illegible]
tax_type_classification	[illegible]	✓	[illegible].
withholding_tax	[illegible]	[illegible]	[illegible].
psid	[illegible]	[illegible]	[illegible].

[illegible] structured/vat-checklists/sa-otp-vat-q3-2025.json [illegible].

```
"checklist": [
  {"n": 1, "label": "Invoice in Arabic language",                    "result": "Y"},
  {"n": 2, "label": "Addressed to Standard Chartered Capital",       "result": "Y"},
  {"n": 3, "label": "Date of issue",                                 "result": "Y",
    "value": "2025-10-30"},
  {"n": 4, "label": "Sequential number",                             "result": "Y",
    "value": "3003093648253002"},
  {"n": 5, "label": "Tax Identification Number (15-digit)",          "result": "NA",
     "regulation": "Supplier is ZATCA itself -- government body"}
],
```

[illegible]

[illegible].

[illegible] chart_field [illegible] 665011-4551-800 [illegible].

[illegible]

[illegible] required_role [illegible] preconditions. [illegible] delegation_of_authority [illegible] x [illegible].

- po-eproc [illegible]
- manual-one-off [illegible]
- manual-recurring [illegible]
- vat-payment [illegible].

[illegible]

[illegible] doi-policy [illegible] activities[] [illegible] controls[] [illegible].

□ □ □ □

□ □ □ □ □ □ □.□ □ □ □ □ □ □ □ □ ocr-result.schema.json.

□ □ □ □ □	□ □ □ □	□ □ □.	□ □ □ □ □ □
ocr_id	□ □ □ □ □ □	✓	□ □ □ □ □ □ □ □ □ □ □ □ □ □ □ □ □ □ □.
source_document	□ □ □ □ □ □	✓	□ □ □ □ □ □ □ □ □ □ □ □ □ □ □ □ □ □

□□□ □□□□□□□□□□□ □□□ □□□□□□□□□□□ □□.□.
□□□ □□□ □□□ □□□□□□□ □□□□□ □□□□□□□□□ □□□
□□□□□□□ □□ docs/schema/common/enums.yaml\#/vat_thresholds
□□□□□□ □□□□ □□ □□□□□□□□□□□□□□ enums.yaml
□□□□□.□□□□□□□□□□

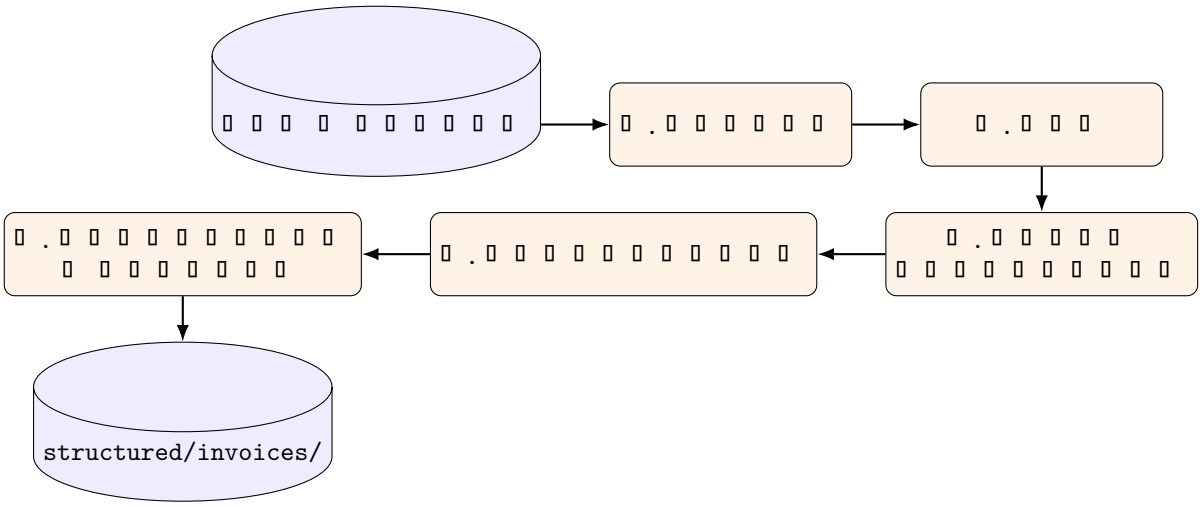
```
python3 docs/schema/validate.py --registry-check
python3 docs/schema/validate.py --domain arabic \
    --schema invoice.schema.json --file invoice.json
```

□□ □□□□ --registry-check □□ □□□□□ □□□□ □□□
□□□□□ □□□□□□□□ docs/schema/ □□ □□□ □□□□□□□□□
□□□□□□□□ □□□□□□.

□□□□□□□□

□□□□□□ □□□□□□□□□□□□□□□□□□□□

□.□ □□□□□□□□□□□□□□□□□□□□


□□□□□□□□□□
□.□□□□□□□
□.□□□□
□.□□□□□□
□□□□□□□□□□
□.□□□□□□□□□□□□
□.□□□□□□□□□□□□
□□□□□□□□□□
structured/invoices/

□□□□□□ □.□□ □□□ □□□□ □□□□□□□□□□□□□□□□□□□□

□ □ □ □ □ □ □ . □ □ □ □ □ □

[illegible]

[illegible] extracted-text/<slug>.txt [illegible] source_document.ocr_text_path. [illegible]

[illegible]

[illegible] invoice.schema.json [illegible] description [illegible]

[illegible]

[illegible]

```
SYSTEM: You extract tax-invoice fields into a JSON object that MUST
```

```
        validate against the provided JSON Schema. If a field is not
        present in the source, omit it -- NEVER invent a value.
USER:   <schema>
        ---
        OCR TEXT (Arabic may appear in RTL order; preserve spelling):
        <ocr_text>
        ---
        Emit JSON only. Set "metadata.language" to "ar", "en", or "ar+en"
        based on the dominant script.
```

[illegible]

[illegible]

[illegible]	[illegible]	[illegible]
electronic_id	[illegible]	[illegible] ^[0-9A-Z]{26}$ [illegible]
	B1STE8→BISTE8 [illegible]	[illegible] --psm 6.
totals.gross	[illegible] ٠١٢٣٤٥٦٧٨٩ [illegible]	[illegible]
vat_rate_percent	[illegible] 0.	[illegible] vat_category [illegible]
signature.kind	[illegible]	[illegible] document_kind [illegible]
seller.name_en	[illegible]	[illegible]

[illegible]

[illegible] → [illegible]

[illegible]

- `seller.name_en` [illegible] `buyer.name_en` [illegible]
- [illegible] `line_items[].description_en` [illegible]
- `seller.address_en` [illegible] `buyer.address_en`.

[illegible]

[illegible]

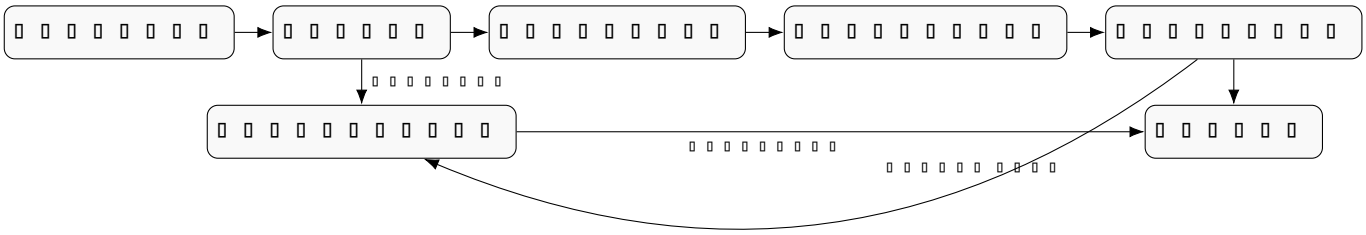
[illegible]

[illegible]	[illegible]
هيئة الزكاة والضريبة والجمارك	[illegible]
بنك ستاندرد تشارترد	[illegible]
ضريبة القيمة المضافة	[illegible]
اشعار صدور فاتورة	[illegible]
فاتورة ضريبية	[illegible]

[illegible] $d = \mathrm{LCS}(a,b)/\max(|a|,|b|)$ [illegible] $d \leq 0.18$ [illegible]

□ □ □ □

□□□□□□. □□□ □□□□□□□□ □□□□□ □□□□□□□ □□□□□
□□□□□□□□□□□ ingest(file)□ retry(id)□ release(id).



□□□□□□ □.□□ □□□□□□□□ □□□□□□□□□□□□□
□□□□□□ □□□ □ □□□□□□ □□□□□□□□□.

□.□ □□□□□□□□□□□□□□□□

□□□□□□ □□□. □□□ □□□□□ □□□□□□ □□□□□□□ □□
structured/invoices/ □□□□ □□□ □□□□□□□□□□□ □□□□□□
□□□. □□□□□ □□□□ □□□□□□ □□□□ □□□□□□□□□□
□□□□□□□□□□□□□□□□ □□□□□□□ □□□□ □□□ □□□□□□
□□□□□ validate.py □□□□ □ jq□ □□□□□□ □□□□□□□□
□□□□□ □□ □□□ □□□□□□□□□.

□□□□□□□□□ □□□□□□□. □□□ □□□□□ □□□□□□ □□
□□□□□□□□□□ □□□□□□□□ □□□ □□□□□□□□□□□□

- □□□□□□□□□□ □□□□□□□□□□□□ □□□□□□□□ □□□□□
  □□□ □□ □□□□□□□ □□□□□ □□ invoice.schema.json□ .
- □□□□□□□□□□ □□ □□ line_items[] □□□□□□□ □□
  □□□□ □□□□□□□□□□□□□□ □□□□□□□ □□□□□
  □□□□□□□□□□□□□□□□□

□□□□□□□□

□□□□□□□□

[illegible]

[illegible] (c, a, p, q, r) [illegible]

- c [illegible]
- a [illegible]
- p [illegible]
- q [illegible]
- r [illegible] `result` $\in$ [illegible] p [illegible].

[illegible] (c, a) [illegible] `checklist[]`. [illegible] `tax_type_classification` [illegible].

[illegible]

[illegible]

[illegible] `result` [illegible].

[illegible]

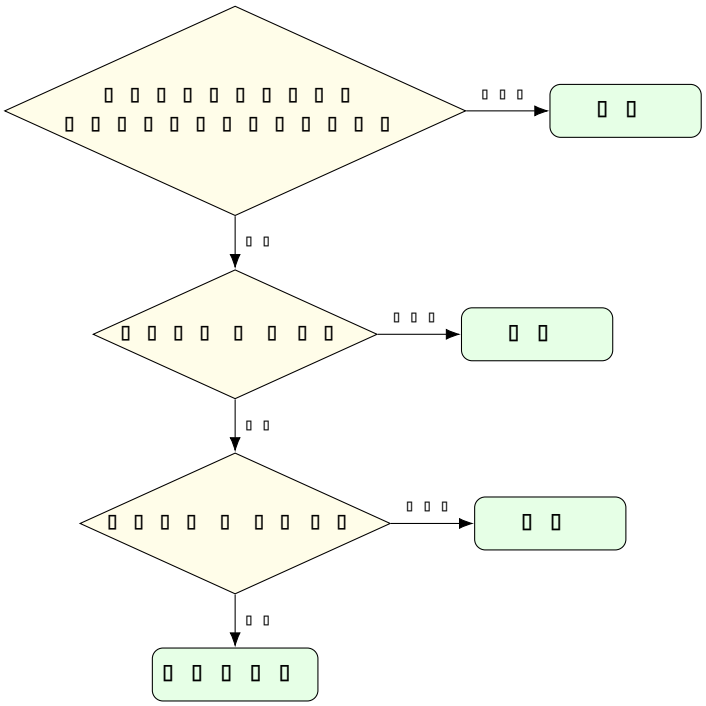
[illegible].

```
if supplier.supplier_type == GOVERNMENT
     and tax_authority == ZATCA:
    classification = RC         # reverse-charge / government payment
elif totals.vat_rate_percent == 0
     and supplier.supplier_type != GOVERNMENT:
```

[illegible]

[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	metadata.language ∈ [illegible]
[illegible]	[illegible]	buyer.tax_id [illegible]
[illegible]	[illegible]	metadata.issue_date [illegible]
[illegible]	[illegible]	metadata.electronic_id [illegible] metadata.invoice_number [illegible]
[illegible]	[illegible]	seller.tax_id [illegible] ^[0-9]{15}$ [illegible]
[illegible]	[illegible]	line_items[].description* [illegible]
[illegible]	[illegible]	totals.vat_rate_percent [illegible] ∈ [illegible]
[illegible]	[illegible]	totals.net [illegible]
[illegible]	[illegible]	totals.gross [illegible]
[illegible]	[illegible]	totals.vat_amount [illegible] net· $r/100$ [illegible]
[illegible]	[illegible]	withholding_tax.applicabl [illegible]

[illegible]



□ □ □ □ □ □ □.□ □ □ □ □ □ □

[illegible] gross [illegible] vat-payment [illegible].

[illegible]

[illegible]

[illegible].

[illegible]	[illegible]
[illegible]	[illegible]
electronic_id [illegible]	[illegible] ^[0-9A-Z]{26}$.
[illegible]	seller.tax_id [illegible] ^[0-9]{9}$.
[illegible]	buyer.tax_id [illegible] ^[0-9]{9}$.
[illegible]	seller.activity_code [illegible].
[illegible]	submission_date − issue_date ≤ [illegible].
[illegible]	[illegible].
[illegible]	[illegible] line_total [illegible] totals.net.

[illegible]

```
if seller.activity_code in EXEMPT_ACTIVITIES:
    classification = EX
elif totals.vat_rate_percent == 14:
    classification = FI
elif totals.vat_rate_percent == 0:
    classification = ZR
else:
    classification = OTHER
```

[illegible] EXEMPT_ACTIVITIES [illegible]. [illegible] 8890 [illegible]. [illegible] 1811 [illegible] totals.vat_rate_percent = 14 [illegible]

[illegible] EGP 661,200 [illegible]

[illegible] **B1STE8RH…SJ10** [illegible] 8890 [illegible] manual-one-off.

[illegible]

[illegible]

[illegible] structured/policies/bh-accounts-payable-doi.json [illegible]

[illegible]

[illegible] vat-thresholds.yaml [illegible] enums.yaml [illegible]

```yaml
thresholds:
  BH:
    si_fi_boundary: 500
    currency: BHD
```

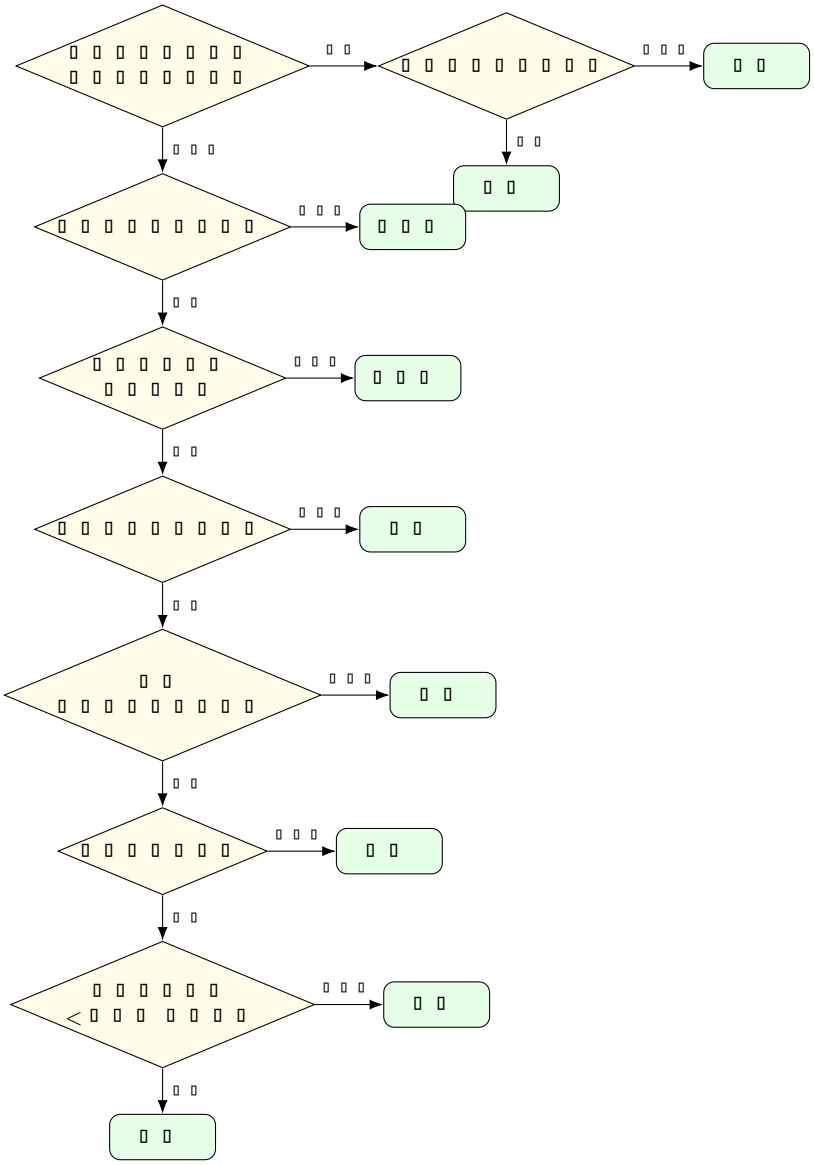
[illegible]

[illegible]

[illegible]

[illegible]

[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]



□□□□□□ □.□□ □□□□□□□ □□□ □□□□□□□

□ □□□□□□□□□□□□□□□□ □□□□□□□ □ □□□ □□□□□
□□□□□□ □□□ □□□□□□□ □□ □□□□□□ □□□□□□□ □ □□□
□□□□ □□□ □□□□□□□□ □□ □□□□□□□□□ metadata.tax_authority
≠ □□□□. □□□ structured/payment-instructions/qa-qcsd-payment-
instruction.json□□□□□□ □□ □□□ □□□□□□□□□□□□.

□.□ □□□□□□□□□□□□□□□

□□□□□□□□□□□ □□□□□ □□□ □□□□□□□□□ □□ □ □□□□
□□□□□□ □□□□□□□ □□□□□ □□ □□□□□□□□□ □□□□□
□□□□□. □□□ □□□□□□□ □□□□□ □ withholding_tax □□□□□
□□□□ applicable□ rate_percent□ □□□ amount. □□□□□□□□□□□
□□□□□□□□□□ □□□□□ applicable = false □□□□□ □□□□□□□
□□□ □□□ □□ □□□□□ □□□□□□□□□. □□□□□□□□□□□□
□□□□□□□ □□□□□□□□□ □□□□□□□ □ □□□□□□□□
□□□□□□□□ □□□□ □□.□. □□ □□□ □□□□□□□□□□ □□□
□□□□□□□□□ □□□□ □□□□ □□ □□□□□□□□□□□□ □□□□
□□□□□□□ □□ enums.yaml □□□□□□ □□□ □□□□□□ □□□
□□□□ □□□.

□.□ □□□□□□□ □□□□□□□□

□□□ □□□□□□ □□□□□□□□ □□□ □□□□□□□ □□□□□□□□□□

- □□ vat-checklist□□□□□□□□□□□□□□ structured/vat-checklists/<slug>.json□
  □□□□□□□□□ invoice□□□□□□ □□□□□ □ □□□□□□□□□□
  totals.vat_category□
- □□ □□□□□□□□ □□ □□□□□□□ □□ □□□□ □□□□□□□□
  □□ □□□ □□□□□□□□□□□ □□□□□□□ □□□□□□□□□□ □□.

□ □□□□□ □□ □□□□□□□□□ □□□□□□□□ □□□ □□□□ □□□
□□□□□□ □.□□□ □□□□□□□ □□□□□ □□□□□□ □□□□□□□□□ □
□□□□ □□ □□□□□□□□ □□□□□□□ □ □□□□□□ □□□□□□□□
□□ □□□ □□□□□□□□□□□□□□□□ □□□□□□□□ □□□□□□□
□□□□.

[illegible]

[illegible]

[illegible]

[illegible]

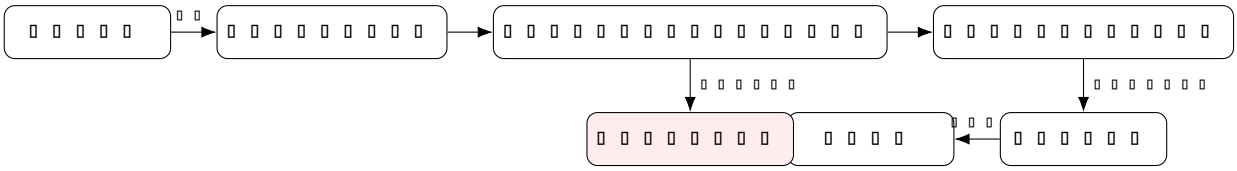
1. [illegible] structured/workflows/[illegible]
2. [illegible]
3. [illegible]
4. [illegible] delegation_of_authority [illegible]
5. [illegible] payment-instruction.schema.json.

[illegible]

□.□ □□□□

[illegible]

[illegible]	[illegible]
[illegible]	[illegible]
[illegible]	[illegible]
[illegible]	[illegible]
[illegible]	po-eproc
[illegible]	manual-one-off
[illegible]	manual-one-off manual-recurring
[illegible]	[illegible]



po-eproc structured/workflows/po-eproc.json .

issue_date.

posted → paid .

. manual-one-off

. manual-one-off structured/workflows/manual-one-off.json .

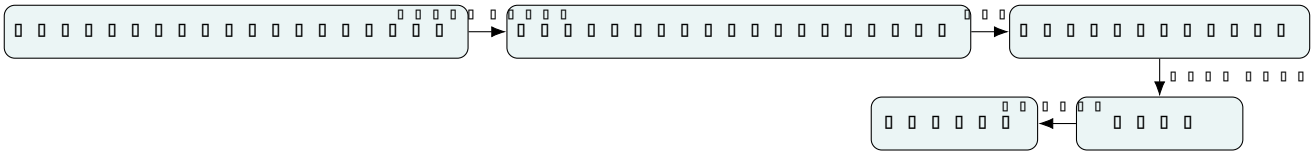
. manual-recurring

dispensation_document_path notes[].

. vat-payment

supplier.supplier_type .

□□□ □□ □□□□ □□□□□□□□□□□ □□□□□ □□□□ □□□□□
□□□□ □□□□ □□□□□□□□□□□□□□□□□□□ → □□□□□□
□□□□□□□□□□□□□□ □□□□□ □□□□□ □□□□□ □□□
□□□□□□□ →□□□□□□□□□□□□ →□□□□ →□□□□□□.



□□□□□□ □.□□ vat-payment□ □□□□□□ □□□□ □□□□□□□ □□
□□□ □□□□□ □□ □□□□ □□□□□□□□□□□. □□□□□ □□□
□□□□□□ □□□□□ □□□ □□□□□□□□□ □□□□ □□ □□□
□□□□□ □□ □ □□ □□.

□.□ □□□□□□□□□□□□□□□□□□□□□□

□□□ □□□□□□□ □□□□□□□ □□□ □□ □ □□□□□□ □□□□
□□ □□□ □□□□□□□□□□ delegation_of_authority □□□□□□
□□□□□□ □□□□□□□□□□ □□ max_amount. □□□ □□□□ □□□
□□□□□□ □□ □□□ □□□□□□ □□□□□□ □□□□□□ max_amount □□
≥ □□□ □□□□□□□□□□ □□□ □□□□□. □□ □□ □□□□□□□
□□□□ □□□ □□□ □□□□□□□□□□□□□□ □□□□ □□ issue_date□
□□ □□□□□□□□ □□□□□□ □□□□□□□□□□□□.

□□□□□ □.□□ □□□□□□□ □□□ □□□□□□□□□□□ □□□
□□□□□□□□.

| □□□□□□□□ | □□□□□□□□ | □□□ □□□ | □□□□□□□□□□□□□□□□□□□□

[illegible] exceptions[] [illegible].

[illegible]

[illegible]

- [illegible] 09-0906743-50
- [illegible] 09-0906743-50

[illegible] >[illegible].

[illegible]

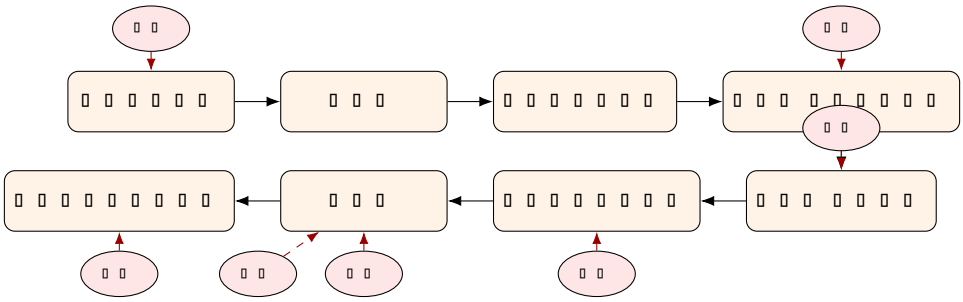
[illegible] (workflow_id, from, to, actor_psid, timestamp, evidence_refs). [illegible] structured/ [illegible] events/ [illegible].

□□□□□□□□

□□□□□□□□□
□□□□□□□□□□□□
□□□□□□□□□□

□.□ □□□□□□□□□□□□□□□□

□□□ □□□□□□□□ □□□□□ □□□□□ □□□□□□□□□ structured/policies/bh-accounts-payable-doi.jsoncontrols[]□ □□□ □□□□□□ □□□□□□ □□ □□□□□ □.□ □□□□ □□□ □□□□□□ □□□□□□ □□□□ □□□□□□□□□□□□ □□□□.



□□□□□□ □.□□ □□□□□□□□ □□□□□□□□□□□□□□ □□□□□□ □□□□□□ □□□ □□□□□□□□□□ □□□□□□□ □□□□□□□□□□□□□□□. □□□□□□ □□□□□□□□ □□□□□ □□□□□ □□□□□□□□ □□□□□□ □□□□□□□□□□□□□□□□

[illegible]

[illegible]	[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]	cfo_approved→[illegible]
[illegible]	[illegible]	[illegible]	[illegible] manual-one-off[illegible]
[illegible]	[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]	[illegible] manual-recurring.
[illegible]	[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]	source_document.sha256.

□□□

```
# Per-field equality of every golden-set record between a reference
# snapshot and the pipeline output.
diff \
  <(jq -S . docs/arabic/structured/invoices/*.json) \
  <(for p in docs/arabic/source/*.pdf; do
      ./pipeline extract "$p"
    done | jq -S .)
```

[illegible].

[illegible]

[illegible].

- [illegible] corpus.json [illegible].
- [illegible] ≥ [illegible].
- [illegible] ≤ [illegible] description_en.

[illegible].

- [illegible] EXEMPT_ACTIVITIES [illegible].
- [illegible].

[illegible].

- [illegible].
- [illegible].
- [illegible].

٤.٤ [illegible]

[illegible] ٤.٤ [illegible] [illegible] [illegible]

[illegible]

[illegible]

1. [illegible]
2. [illegible]
3. [illegible]
4. [illegible] CFO [illegible] vat-payment [illegible] manual-one-off [illegible].

[illegible]

[illegible].

[illegible].[illegible].

[illegible].[illegible].

[illegible]. [illegible] events/[illegible]

[illegible]	[illegible]	[illegible]
[illegible] → [illegible]	[illegible]	[illegible]
[illegible] → [illegible]	[illegible]	[illegible]

[illegible] structured/corpus.json [illegible]

[illegible]

[illegible]

[illegible]

[illegible] workflow_id, from, to, actor_psid, timestamp, evidence_refs, [illegible]

- [illegible]
- [illegible]
- [illegible]

[illegible]

[illegible]

[illegible] structured/corpus.json [illegible] .

[illegible]

[illegible]	[illegible]	[illegible]
[illegible]	docs/arabic/source/3CPJR5CQ4PNNN9A4XHBB370K10.pdf	2b912687fa94…
[illegible]	docs/arabic/source/B1STE8RHNWZY9TYEXDS6WSZJ10.pdf	18eb35671131…
[illegible]	docs/arabic/source/اشمارصدورفاتورة [illegible]	cbbb3f702ad5…
[illegible]	docs/arabic/source/EDAAAnnualFeeInvoice2025.pdf	cf91c2a43574…
[illegible]	docs/arabic/source/Instructionforpaymenttovendor(QCSD).pdf	3d9d62690d0f…
[illegible]	docs/arabic/source/Saudi-OTPEntryprocessing-VATQ3'25(1).docx	3a6c8f37528f…
[illegible]	docs/arabic/source/AccountsPayableDOI.docx	128ff6792798…
[illegible]	docs/arabic/source/FW-AI_ArabicInvoice...eml	e01ba78d5ca5…

□.□ □□□□□□□□ □□□□

□□□□□ □□□□□□□ □□□ □□□ □ □□□□□ □□□ □□□□□□□□□□ extracted-text/□ □□□□□□□□ □□□□ pdftoppm -r 300 | tesseract -l ara+eng. □□□□□□□ □□□□ □□□ pdftotext -layout□ □□□□ □□□□□□□□□ □□□ pandoc -t plain□ □□□ □□□□□□□ □□ □□□ □□□□□□□□ □□ □ □□□□□ □□□□□□ □□□□□□ □□□□□ □□ □□□□□□□□□□□□□□□□□□□□□□□□□ email□□□□□□.

□.□ □□□□□□□□□□□□□□□□□□□□

□□□□□□□□□□□□□□□□□□□

□.□ □□□□□

□□□□	□□□□□□□
pdftoppm	□□□□□□□□□□ □□□ □□□□□ □□ □□□ □□□ □□□□□□□□□.
tesseract	□□□□□□□□□□ □□□ □□□□ -l ara+eng.
pdftotext	□□□□□□ □□□□ □□□ □□□□□□□□ □□□ □□□□□□□□□□□□□ □□□ □□□□□ □□□□.
pandoc	□□□□□□□ .docx □□ □□□□□ □□□□.
□□□□□□ email	□□□□□ □□□ .eml □□□ □□□□□□ □□□ □□□□ □□□□□.
jq	□□□□□ □□□ □□□□□□□ □□□□□ corpus.json.
jsonschema □ referencing	□□□□□□□ □□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□□□□□□□□□□□□□□□.
xelatex □ tectonic	□□□□□ □□□□ □□□□□□□□□□□□□□□□□□□□

□ □ □ □ □ □ □ □ .□ □ □ □ □ □ □ □ □ □ □ □ □ □ □ □ □ corpus.json□
□ □
□ □ □ □ □ □ □ □ □ □ .

```
{
  "schema_version": "1.0",
  "items": [
    {
      "id": "eg-eta-3CPJR5CQ4PNNN9A4XHBB370K10",
      "type": "invoice",
      "country": "EG",
      "tax_authority": "ETA",
      "sha256": "2b912687fa94...",
      "source_path":     "docs/arabic/source/3CPJR5CQ...K10.pdf",
      "text_path":       "docs/arabic/extracted-text/eg-eta-...K10.txt",
      "structured_path": "docs/arabic/structured/invoices/eg-eta-...K10.json"
    },
    {
      "id": "sa-zatca-ishaar-005",
      "type": "invoice",
      "country": "SA",
      "tax_authority": "ZATCA",
      "sha256": "cbbb3f702ad5...",
      "source_path":     "docs/arabic/source/<arabic-name>.pdf",
      "text_path":       "docs/arabic/extracted-text/sa-zatca-ishaar-005.txt",
      "structured_path": "docs/arabic/structured/invoices/sa-zatca-ishaar-005.json"
    },
    {
      "id": "qa-edaa-annual-fee-2025",
      "type": "payment_instruction",
      "country": "QA",
      "tax_authority": null,
      "sha256": "cf91c2a43574..."
    }
  ]
}
```

□.□ □□□□□□□□□□□□□□□□□□□□□□□□

□□□□□□□□□□□□□□□□□□□

[illegible]

[illegible]

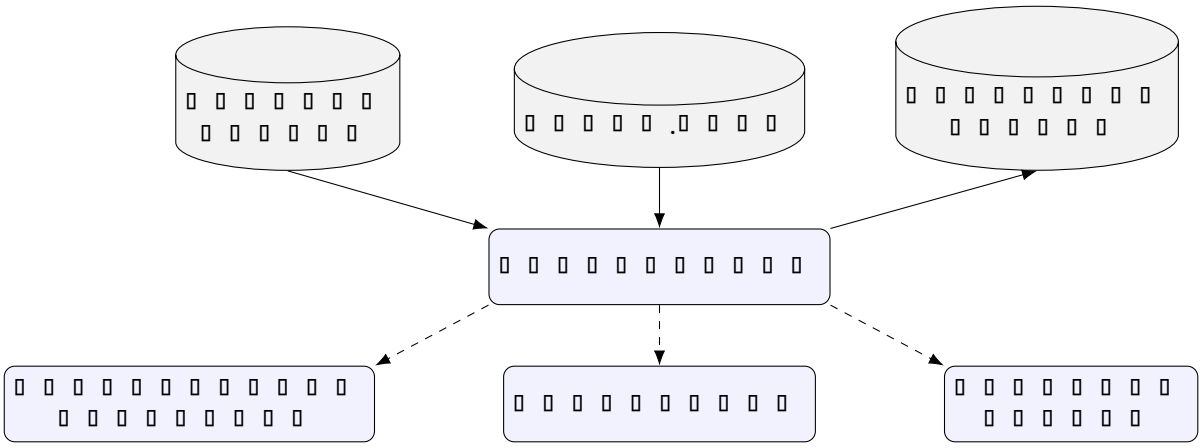
[illegible]

[illegible]

[illegible]

[illegible]

1. [illegible]
2. [illegible]
3. [illegible]
4. [illegible]



□□□□□□□ □.□□ □□□ □□□□□□□ □□□□□□□□□ □□□□□□□□□

□ □ □ □ □ □ .□ □ □ □ □ □

```
        - source: str  # regulation reference
    """
```

□ □ □ □ □ □.□ □ □ □ □ □

□.□.□ □□□□□□□□□ □□□□□

□□□□□□□□ □□□□□ □□□ □□□ □□□□□□□□□□ □□ □□□□□□ □□□□□□ □□ □□□ □□□□□□□□ □□□□□□. □□□ □□□□ □□□□□□□□□ □□ □□□ □□□□□□□□□□□□ □□□□□□□□□ □□□□□ □□□□□□□ invoice.schema.json.

□□□□□□□ □.□□ □□□□□□□ □□□□□□□□ □□□□□□□□ □□□□□ □□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□□ □□□□.

```
totals.gross           # Invoice gross amount
totals.vat_rate_percent  # VAT rate as percentage
seller.tax_id          # Seller VAT registration number
seller.country         # Seller country code
buyer.country          # Buyer country code
metadata.issue_date    # Invoice issue date
line_items.0.description  # First line item description
```

□.□.□ □□□□□ □□□□□□ □□□□□□□□□□□□□□

□□□□ □ □□□□□□□□□□ □□□□□□□□ □□□□□□□ □□□□□ □□□□□□□□□□□□□□□

□ □ □ □ □ □ □.□ □ □ □ □ □ □

.

[illegible]

[illegible]

[illegible]

[illegible] ≥ [illegible]

[illegible]

[illegible] .

[illegible] .

[illegible] .

[illegible]

[illegible]

[illegible]

[illegible]

[illegible] .

[illegible]

[illegible] .

[illegible]	[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]	
[illegible]	[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]	[illegible]

[illegible]

[illegible] tests/fixtures/arabic/.

[illegible]

[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]

[illegible]

[illegible]

[illegible]

- [illegible]
- [illegible]
- [illegible]
- [illegible]

[illegible]

[illegible]

[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]

□.□.□ □□□ □□□□□□□□□□□

□□	□□□□□□□□	□□□□□□□□□□□□
□□□□□□□□□□□	□□	□□□□□□□□□ □□□□□□□□□□□□

□□□□□□□□□ □□□□□□□□□□ □□□□□□□□□□

□.□ □□□□□□□□□□□□□□□□□□□□□□□□

□□□ □□□ □□□□□□□□ □□ □□□□□ □.□ □□□ □□□□□□□□
□□ □□□ □□□□□□□□□□□□ □□□□□□□□□ □□□□□□□□ □□ □□□□
□□□□ □.□.□ □□ □□□□□□□□□□ □□□□ □□□□□□□□ □□ □□□□
□□□ □□□ □□□□□□□□□□□□ □□□□□□□□□□ □□□ □□□□
□□□ □□□□□□□□ □□□ □□□□□□□ □□□ □□□□□□ □□
□□□□ □□□□ □□□ □□ □□□□□□ □□□□□□□□□□□□□□ □□□□
□□ □□□□□□□□□□□□ □□□□□□□□□□.

□□□□□ □.□□ □□□□□□□ □□ □□□□□□□□□□□□□□ □□□□□□
□□ □□□□□□□□□□□□ □□□□□□□□□□.

□□□□□□□□□□	□□□ □□□□□□□	□□□□□□□□□□□	□□□□□
□□□ □□□□□□□□□□□	□□□□□□□	□□□□□□	□□□ □□□□
□□□□□ □□□□□□□□□□□□□□ □□□□□	□□□□□□□	□□□□□□	□□□ □□□ □□□□□
□□□ □□□□□ □□□□□□□□□□□	□□□□□□	□□□□□□□□□□	□□ □□□ □□□□ □□□ □□□
□□□□ □□□□□□□□□□□ □□□	□□□□□□□□	□□□□□	□□□□□□□□□ □□□ □□□□□

□.□.□ □□□□□□□□□□□□□□□□□□□□□□□□

□□□.□ × □□□□□□□ □□□□□ □□□□□ □□□□□ □□□ □□□□□ □□□□.
□□□□□□ □□□□□ □□□□□□□□ □□□□ □□□□ □□□□□□□□□□□□□□
□□ □□□□ □□□□□□□□□□□□ □□□□ □□□□□□□□□□□□□□□□□□□
□□□□□ □□□□□□□□.

□□□ □□□□□□□.□□□□□□□□□□ □.□.□ □□□□□ □□□□□□
□□□□□□□□ □□□□□ □ ara□ ara_old□ □ □□□□□□□□□. □
□□□□□□ □□□□□□□□□ □□ □□□□□□□ □□ □□□□□ □□□□□□
□□□□ □□□□□□□□□□□ □□□□.

□□□. Qwen2.5-14B-Instruct □□□□□□ □□□ □□□□□ □.□.□□
max_model_len=8192□ □□□□□□□□□□□□□□□□□□□□□□□□□□
□ ∼□ □□□□□ □□□□□□ □□□□□□□□ ∼□□□ □□□□□□□ □□□
□□□□□.

□□□ □ □□□□.□ □□□□□□ □□□□ □□□ □□□□ □□□□.

□□□□□□□.□□□□□ □□□□□ □□□□□□□□ □□□□□ □□□□□□□□
□□□□ □□□□□□□ □□□□□□□□ □□□□□ □□□□□□ □□□□.

[illegible] .[illegible] [illegible] <[illegible] .[illegible] [illegible] .[illegible] [illegible] [illegible] [

□.□ □□□□

[illegible]

[illegible] $\rightarrow$ [illegible] $\leq$ [illegible].

[illegible]

[illegible]

1. [illegible]
2. [illegible]
3. [illegible]

[illegible]

[illegible] $\rightarrow$ [illegible]

[illegible]

[illegible]

[illegible]

[illegible]	[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]	[illegible]

[illegible]

1. [illegible] >[illegible] <[illegible]
2. [illegible]

[illegible]

[illegible]

1. [illegible]
2. [illegible] $\kappa > 0.9$[illegible] $\leq$[illegible]
3. [illegible] .[illegible]
   [illegible]
   [illegible] pilot/disputes/<doc-id>.json.
   [illegible]
4. [illegible] pilot/ground-truth/<country>/<doc-id>.json[illegible] docs/schema/arabic/ocr-result.schema.json

□ □ .□ □ □ □

```
    low_threshold = find_accuracy_floor(sorted_data, floor=0.50)

    return high_threshold, low_threshold
```

[illegible] **target_precision = 0.95**. [illegible]

[illegible]

[illegible] → [illegible] ≤ [illegible]

[illegible] → [illegible] → [illegible]

□. □□□□□□□ □□□ □□□□□□□□□□□□ □□□□□□□□□□□□

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]

[illegible]

□□□ □□□□□□ □□□ □□□□ □□□□□□ □□□ □□.□□□ □□□ □□□□□ □□□□□□ □□□□□□□□□□□□□ □□□□□□ □□□□□□ □□□□ □□□□□□□□□ □□□□□□□□□□□□□□ □□□□□□□□□□□□□□ □□ □□ □□□□□□□□□□□□□□□□□□□□□□□□□□□□ □□□□ □□□□ □□□□□□□□□ □□□ □□□□□□□□ □□□□□□□□□□□□ □□□□ □□□□□□□□□□□□□□□□ □□□□ □□□□□□□□□□□□□□□□□□□□

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

□□□□ □□□ □□ □□□□ □□□□□□□□□□□□□□□□.

□□.□ □□□□□□□□□□□□□□□□□□□□

□□□□□ □□□□□□□□□□□□ □□□□□ □□□□□□□ □□□ □□ □□□□ □□□□□□□□□□□ □□□□□□□□□□□□□□□□□□□□

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]	[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]	[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible] <[illegible]

[illegible] >[illegible]

[illegible] >[illegible]

[illegible] reserve/complete [illegible]

[illegible]

□□□□□ □□.□□ □□□□□□□□□□□□□□□□ □□□□□□□□□□□□.

□□□□□□□□□□ □□□

□□□□□ □□□□□□□□□□□□□□□□

[illegible]

[illegible]

[illegible]

- [illegible] ≥ 15 [illegible]
- [illegible]
- [illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

  - [illegible] $\geq$ 20 [illegible]
  - [illegible] $\geq$ 30 [illegible]
  - [illegible]
  - [illegible]

[illegible] `docs/arabic/business-case/refresh-history.yaml`.

[illegible]

[illegible]

[illegible]

[illegible]. [illegible]

[illegible]

□□.□ □□□□□ □□ □□□□□□□□□□□□ □□□ □□ □□□□□□

□□ □□□□□□ □□□□□□ □□□ □□□□□□□□□□□□ □□□ □□□□□□□□ □□□ □□□□□□□□□□□□□□□□ □□□□□□ □□□□ □□□□□□ □□ □□□ □□□□□□□□□□ □□□□□ □□□□□□ □□□□□□

- □□□□□□□ □□□□□□□□□□□□ □□□□□□□□□□□. □□□□□ □□□□ □□□□□□ □□ □□□□□□□ □□□□□□□□. □□□ □□□ .secrets.example □□□□□□□ □□□ □□□ □□□ □□□□□ □□□□□□□□.
- □□□□□□□ □□□□□□□□ □□□□□□□□□□□. □□□□□□□□□□ □□□□□□□□ □□□□□. □□□□□□□□ □□□□ □□□ □□□□□□□ □□□□□□□□ □□□□ □□□□□□□ □□ □□□□□□□□□ □□□ □□□□□ □□□□□□□.
- □□□□□□□ □□□□□□□□□□□□ □□□□□□□□□□□□□□□. □□ □□□□ □□ □□□□□□□□□ □□□□□□□□ □ □□□□□□ □□□□□□□□ □□□□□□□□ □□ tests/fixtures/ □□□ □ □□□□□□□ □□□□□□□□□□□□□□□ □□□.
- □□□□□□□ □□□□□□□□□□□□□□□□□□. □□ □ □□□□□□□ □□□□ □□□□□□□ □□□□□□ □□□ □□□□□ □□□□ □□□ □□□□□□ □□□ □□□□ □□□□□ □□□□□□□□□□□ □□□□ □□. □□□□□□□ □□□□□□□ □□□□□□□ □□□ □□□□□□□□□□□ □□□□□□□□□□□□□□□□□.
- □□□□□□□ □□□□□□□ □□□□□□□□□□□□□. □□□□□□□ □□□□□□□□ □□□□□ □□□□□ □ .clinerules□ □□□ □□□□ □□□□□□□□ □□.□.□ □□□□□ □□□□□□□□□ □□□□□□□□ □□□□□□□□□□□□□□ □□□□□ □□□□□□ □□□□□□□□□□□ □□□□□□□□□□□□□□□.

□□.□ □□□□□ □□□□□□□□□□□□□□ □□□□□□□□□□ □□□□ □.□□□□□□□□□□□□

□□□ □□□□□□□□□ □□□□□□□□□ □ □□□□□□□□□□□ □□□ □□□□□□□□□ □□□□□□□□□ □□□ □□□□□ □□□□□□□□□.
□□□□□□□□□□□□□□□ □□□□□□□ □□□□ □□□□□ □□□ □□□□□ □□□□□□□□□□□□□□□□□□

[illegible]

1. [illegible] `/.clinerules` [illegible]
2. [illegible] `/src/intelligence/.clinerules` [illegible]
3. [illegible] `/src/<domain>/.clinerules` [illegible]

[illegible]

[illegible]

[illegible]

[illegible]	[illegible]	[illegible]
deny_by_default	true	[illegible]
require_identity	true	[illegible]
max_retries	3	[illegible]
timeout_ms	30000	[illegible]

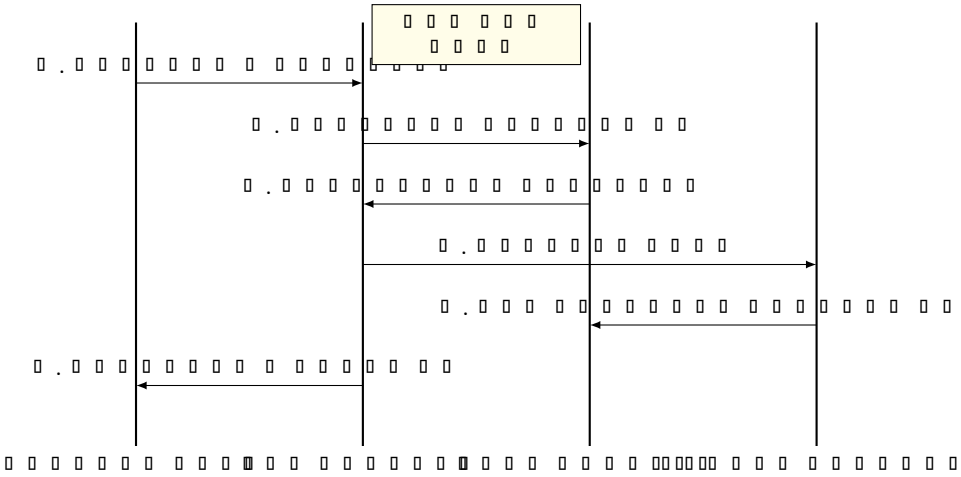
[illegible]

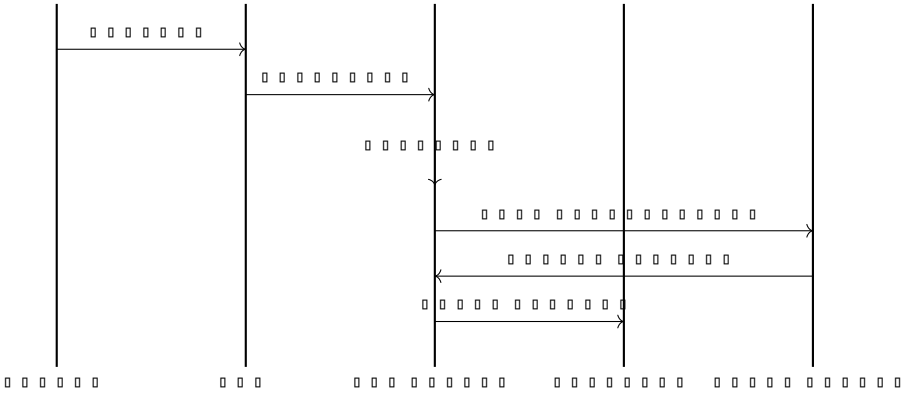
[illegible]

[illegible]

> [illegible]
>
> [illegible]

> □□□ □□□□





[illegible] 1. [illegible]

[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]

[illegible] 1. [illegible]

[illegible]

- [illegible]
- [illegible] 3σ [illegible]
- [illegible]

[illegible] 2. [illegible]

[illegible]

> **□□□

- [illegible]
- [illegible]

[illegible]

[illegible]

- [illegible]
- [illegible]
- [illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]

[illegible]
[illegible] structured/schema/ [illegible].
[illegible].

[illegible] [illegible]

[illegible]. [illegible] docs/latex/specs/arabic/ [illegible]
[illegible] structured/schema/ [illegible] [illegible].
[illegible]
[illegible] src/intelligence/ai-core-pal/data_products/registry.yaml [illegible]
[illegible] arabic [illegible]
[illegible] confidential [illegible] [illegible] vllm-only [illegible]
[illegible]. [illegible] requirement.schema.json [illegible]
[illegible].

[illegible] [illegible] [illegible] [illegible] [illegible]

[illegible] ap-clerk.
[illegible]

[illegible].
- [illegible] po-eproc [illegible] direct-ap [illegible] [illegible].
- [illegible] /ap/{tenant}/queue [illegible] [illegible] ap-clerk.
- [illegible] invoice.intake، invoice.ocr، invoice.extract-fields، invoice.translate-ar-en، vat.validate، vat.calculate، ap.workflow.advance، ap.workflow.park. [illegible].

[illegible]

[illegible]

1. .clinerules [illegible]
2. src/intelligence/.clinerules [illegible]
3. src/arabic/.clinerules [illegible]

[illegible]

[illegible].

[illegible]

[illegible]

> [illegible]
>
> [illegible]

[illegible]

[illegible]

```
// Verify Task A1
{
  "test_fixture": "tests/fixtures/arabic/invoice_sample_01.png",
  "expected_outcome": "ocr_confidence > 0.88",
  "validation_command": "make test-arabic-ocr"
}
```

[illegible] `make spec-arabic` [illegible]

[illegible]

[illegible]

[illegible]

[illegible]. [illegible]. [illegible] ap-clerk.

[illegible]

[illegible]. [illegible] [illegible]

[illegible]

[illegible]. [illegible]

[illegible] . make spec-arabic [illegible] .

[illegible] . [illegible]

[illegible] .

- [illegible] docs/latex/specs/regulations/chapters/03-mgf-framework.tex [illegible] .
- [illegible] 10-identity-attribution.tex [illegible] .
- [illegible] 08-monitoring-spec.tex [illegible] .

[illegible].

- [illegible] 06-conformance-tooling.tex [illegible].

[illegible]

[illegible]

[illegible] structured/schema/ [illegible].

[illegible]

[illegible] invoice.extract-fields [illegible] invoice.schema.json [illegible].

[illegible]

[illegible].

□□□ □□□□□□□□□□□ □□□□□□□□□□ □□□□□ □□□□□ □□□□□□□ □□□□□□□ □□□ □□ □□□ □□□□ □□□ □□□□□□ □□□ □□□□□□□□□ □□□□□□□□□□□

[illegible]

[illegible]

[illegible]

[illegible]

1. [illegible]
2. [illegible]
3. [illegible] $ref [illegible]
4. [illegible] requirement.schema.json [illegible]
5. [illegible]
6. [illegible]

[illegible] .

[illegible]

[illegible] .

[illegible]

- [illegible] docs/latex/specs/tb/chapters/12-implementation-instructio
- [illegible] docs/latex/specs/regulations/chapters/12-implementation-inst
- [illegible] docs/latex/specs/simula/chapters/12-implementation-instructions.tex.

[illegible]

- [illegible] src/intelligence/ai-core-pal/data_products/registry.
- [illegible] src/intelligence/schema_pipel
- [illegible] src/training/pipeline/.
- [illegible] generativeUI/training-webcomponents-ngx.

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

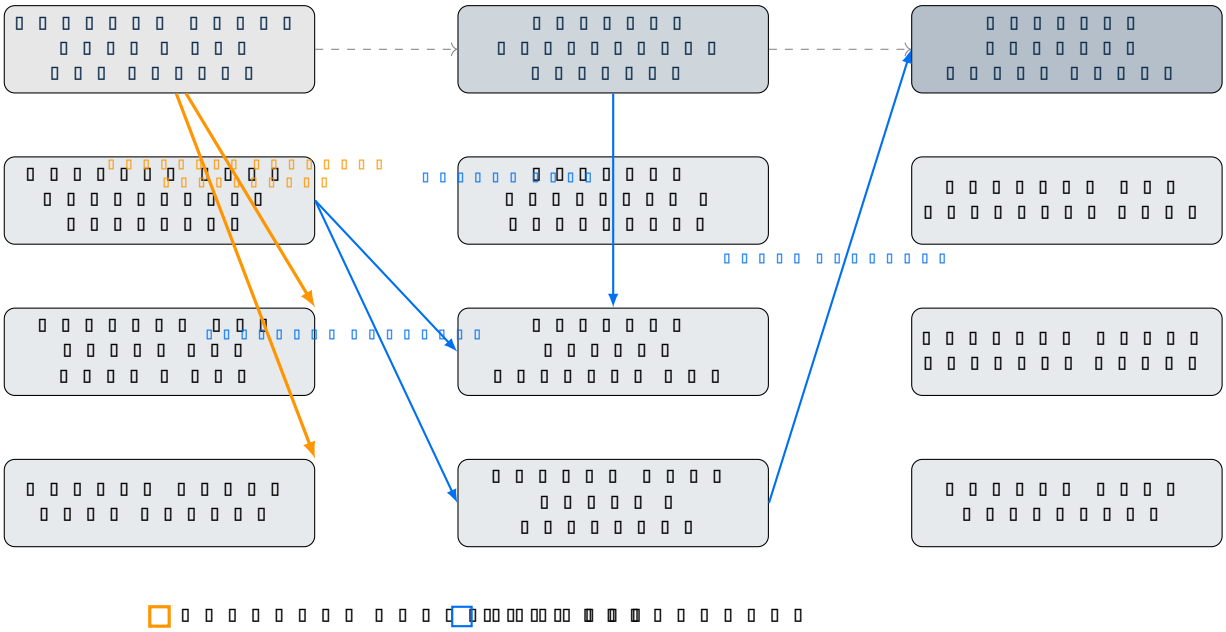
[illegible]

> [illegible]
>
> [illegible]

[illegible]

[illegible]	[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	REG-IDENTITY-01	[illegible]
[illegible]	[illegible]	SIM-EXTRACT-01	[illegible]
[illegible]	[illegible]	SIM-REGISTRY-01	[illegible]
[illegible]	[illegible]	REG-MGF-P1	[illegible]
[illegible]	[illegible]	ARA-AP-POST	[illegible]

[illegible]



□ □ □ □